

Background

Problem: Combustion of fuel produces harmful greenhouse gases, causing climate change



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Solution: Hydrodeoxygenation of Biomass



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How can HDO be scaled industrially?

Biomass: renewable organic materials, which have reduced emissions when burned.

Catalytic Hydrodeoxygenation (HDO): removing oxygen from a compound to make it suitable as fuel, sped up with a catalyst.

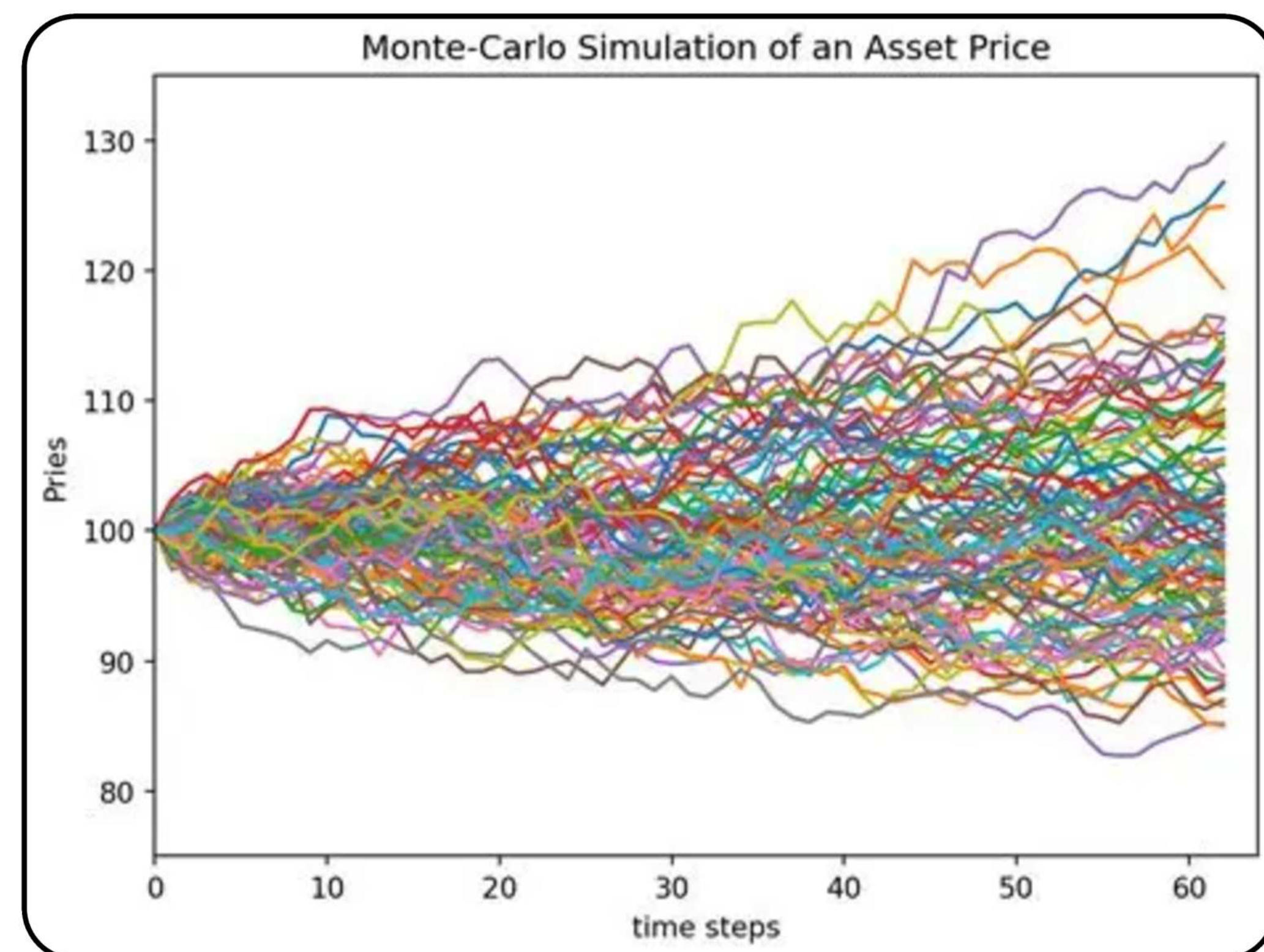
Previous Studies

Palladium-based catalysts showed promising conversion rates (Singh et al.). However, methods previously used with palladium are hard to scale.



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Economic Analysis



(Ryckel & Francois, 2023)

Monte-Carlo Simulation: A Monte-Carlo simulation performs many, many trials to create predictions, even when there are many changing or unpredictable variables.

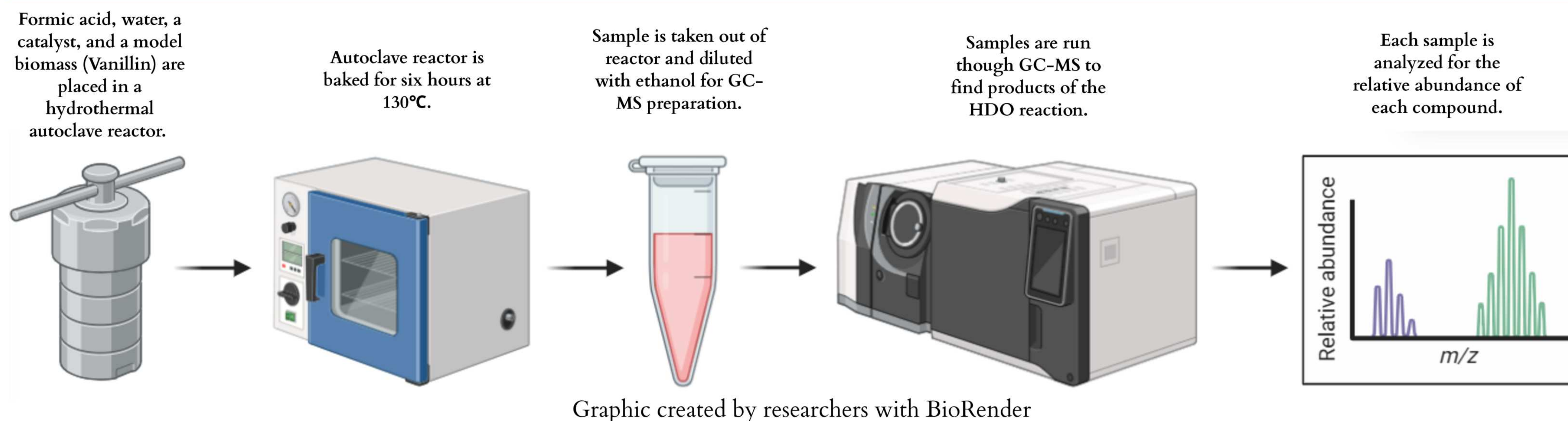
Research Goal

This experiment aims to identify which of the selected catalysts produces **the highest yield** through a hydrodeoxygenation reaction and if they are **economically viable**.

The Economic Viability of Selective Hydrodeoxygenation of Vanillin Using Various Palladium-based Catalysts

Methodology

Hydrodeoxygenation & GC-MS



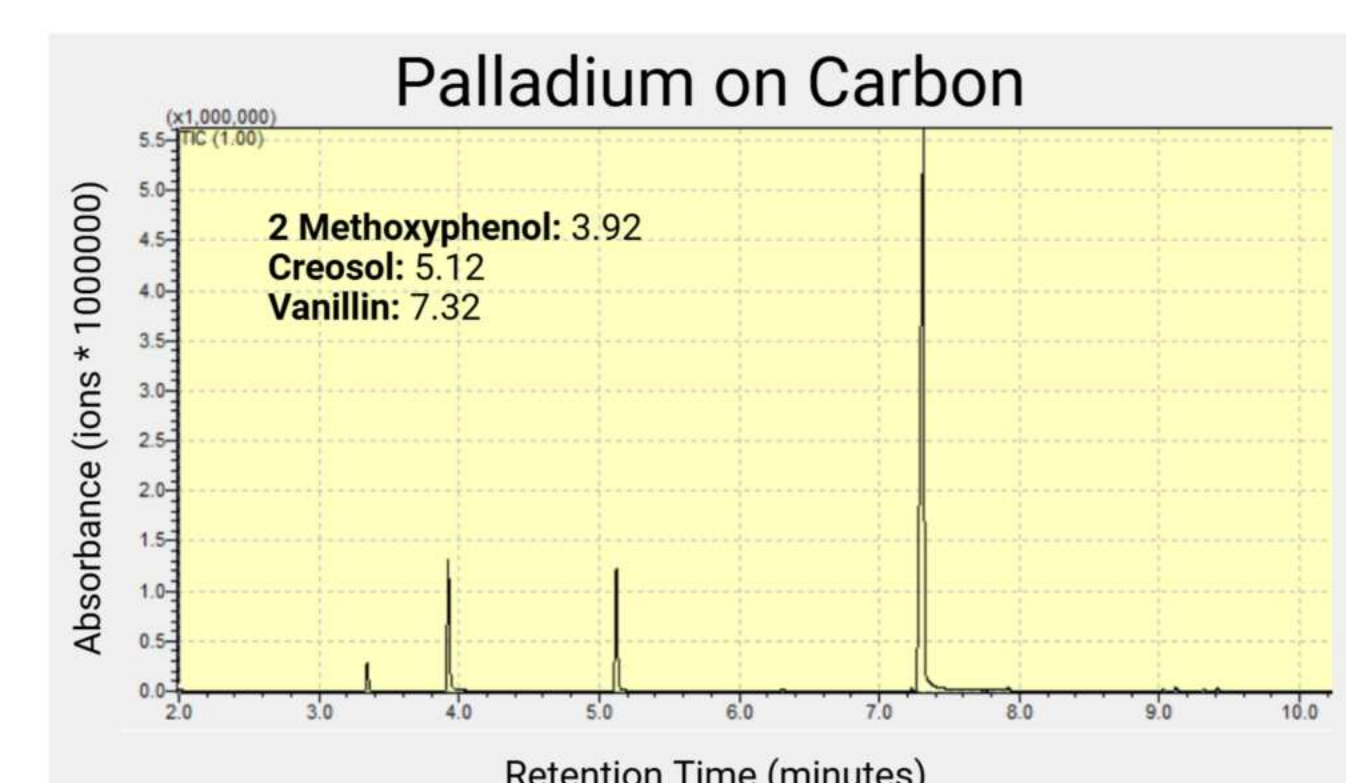
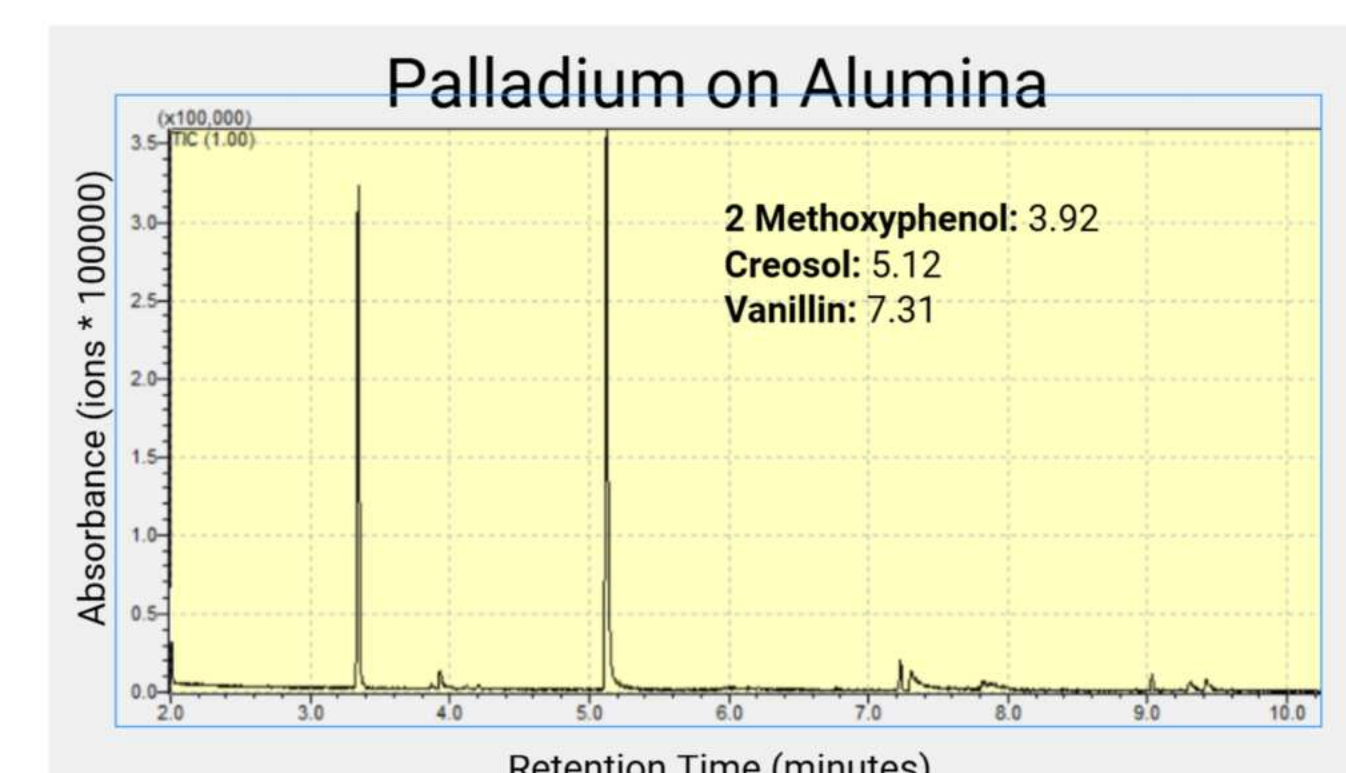
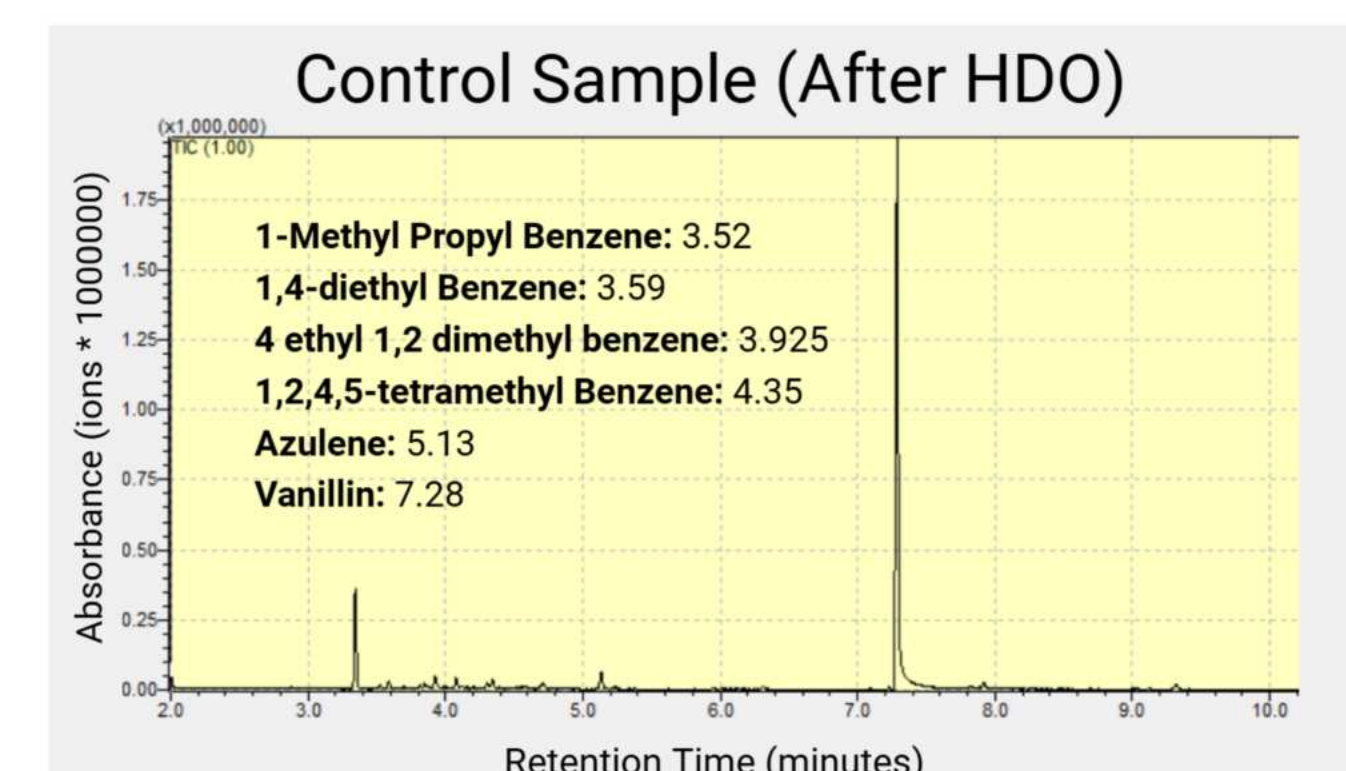
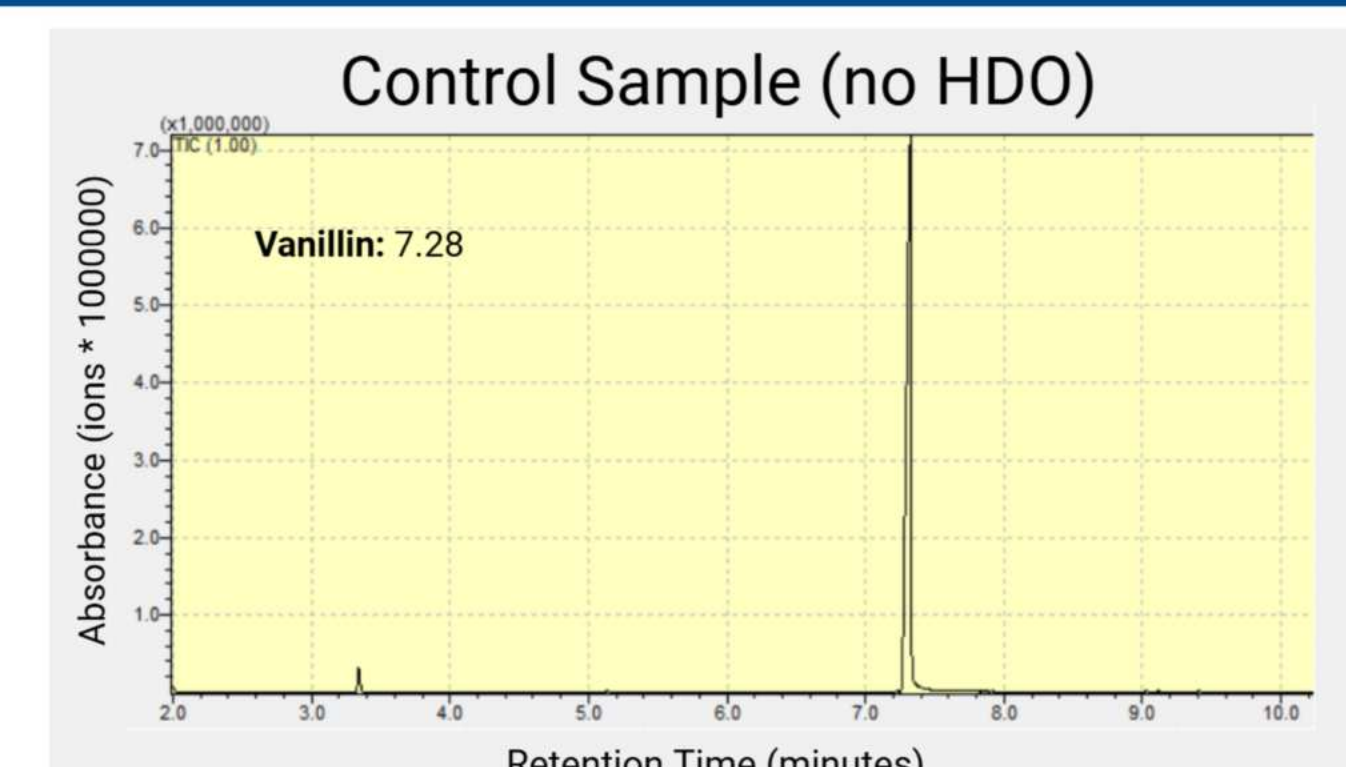
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Economic Analysis

Conversion rates for each catalyst were inputted into their respective Monte-Carlo Simulation. The variable x (factor x) represents a randomized multiplier, which is $\pm 25\%$. The predicted break-even point is calculated with the following equation, which was simulated 5,000 times for each catalyst:

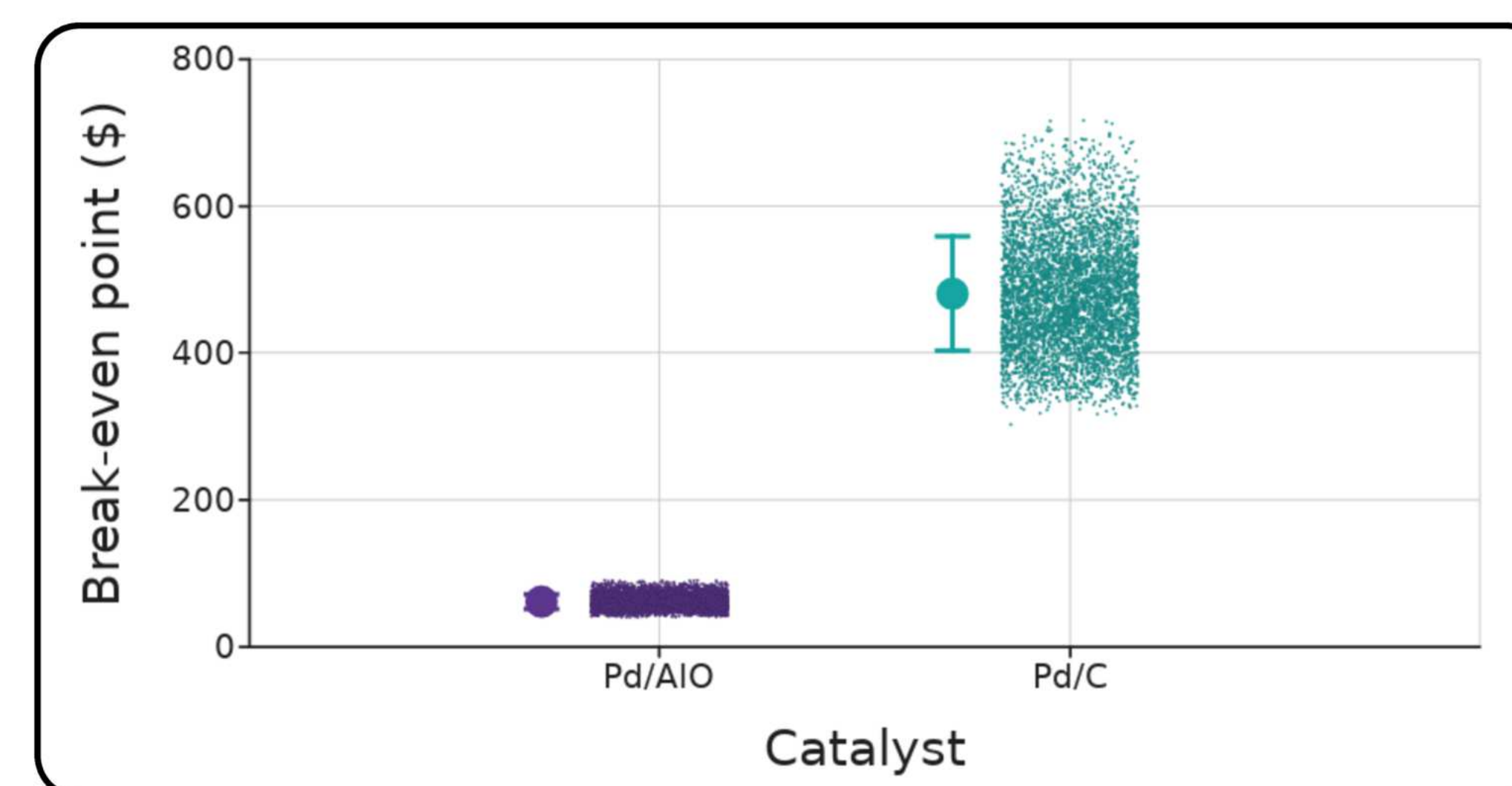
$$\text{Break-even point (\$)} = \underbrace{([P_{va} \cdot Q_{va} \cdot x])}_{\text{Cost associated with vanillin}} + \underbrace{([P_{fa} \cdot Q_{fa} \cdot x])}_{\text{Cost associated with formic acid}} + \underbrace{([P_w \cdot Q_w \cdot x])}_{\text{Cost associated with water}} + \underbrace{([P_{ca} \cdot Q_{ca} \cdot x])}_{\text{Cost associated with catalyst}} \cdot \underbrace{\frac{1000}{1.1}}_{\text{Conversion factor to 1 Lier}} \cdot \underbrace{\frac{1}{R_{con}}}_{\text{Conversion factor of yield}} + \underbrace{[P_e \cdot Q_e \cdot x]}_{\text{Cost associated with energy use}}$$

Results



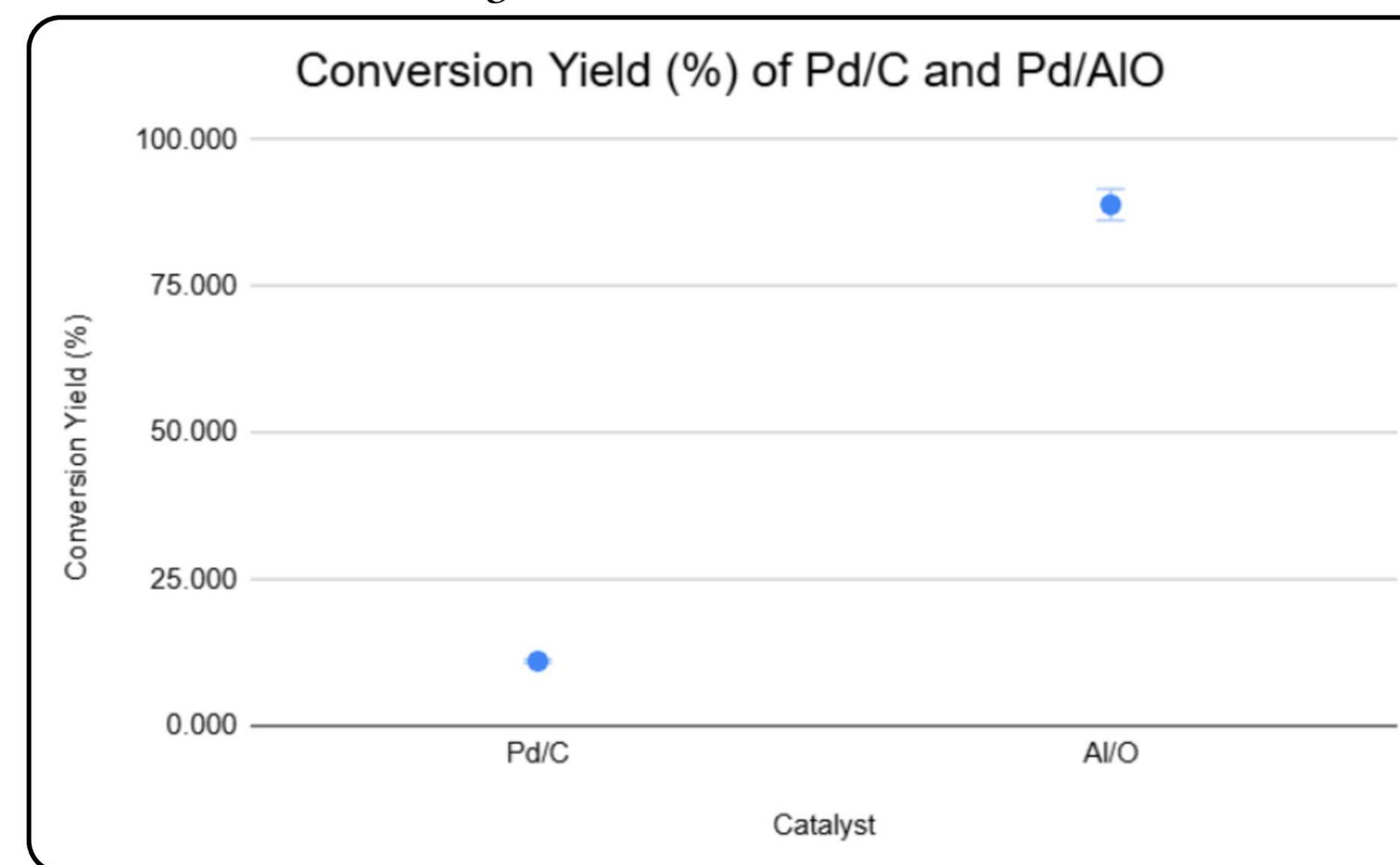
Graphs created by Researchers using Shimadzu GC-MS

Palladium on Carbon Has a Higher Break-Even Point with More Variability



Graph created by researchers with DataClassroom

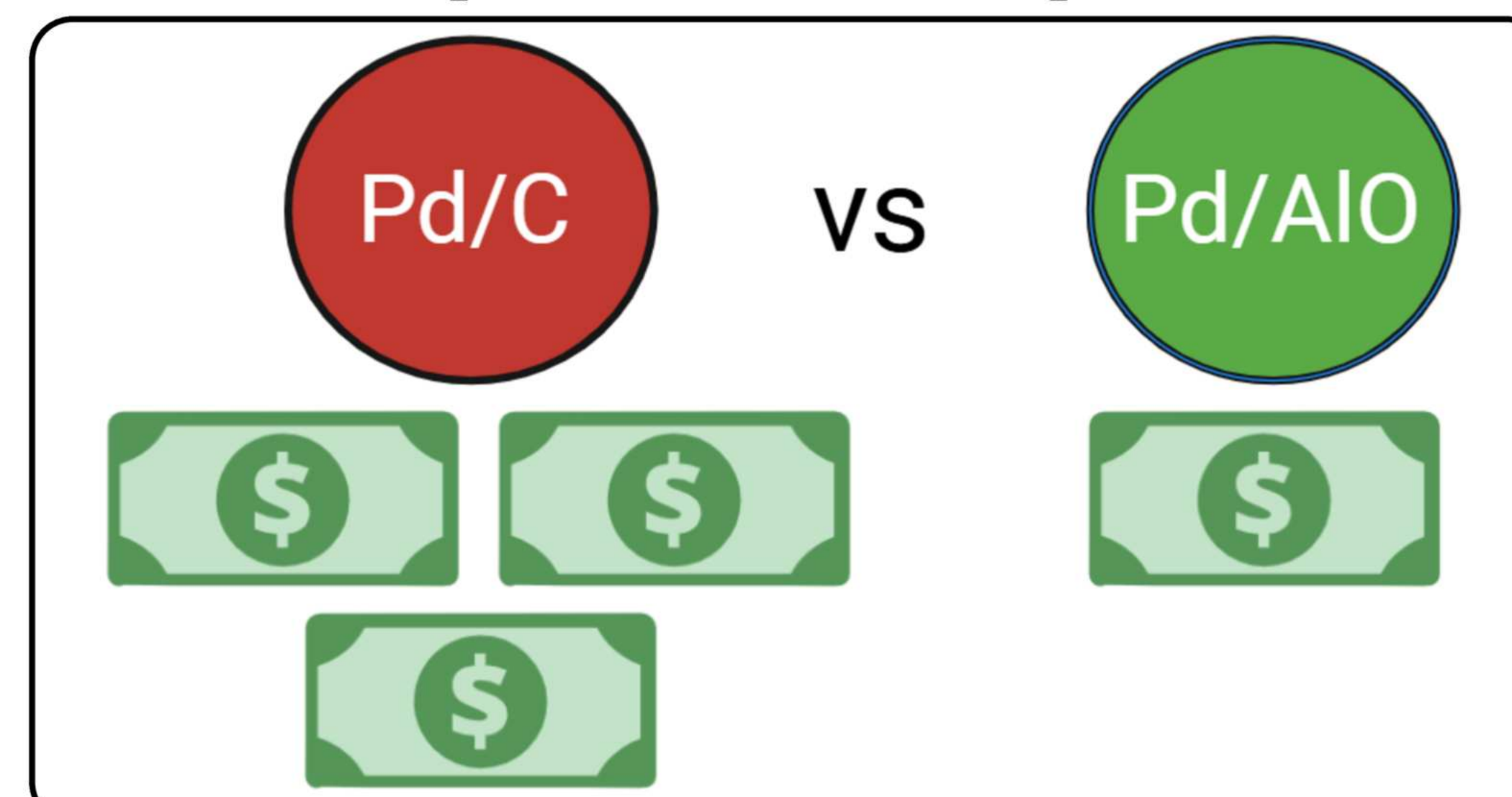
Palladium on Carbon is Has a Higher Conversion Rate of 88% vs Palladium on Alumina of 11%



Graph created by researchers with Google Sheets

Discussion

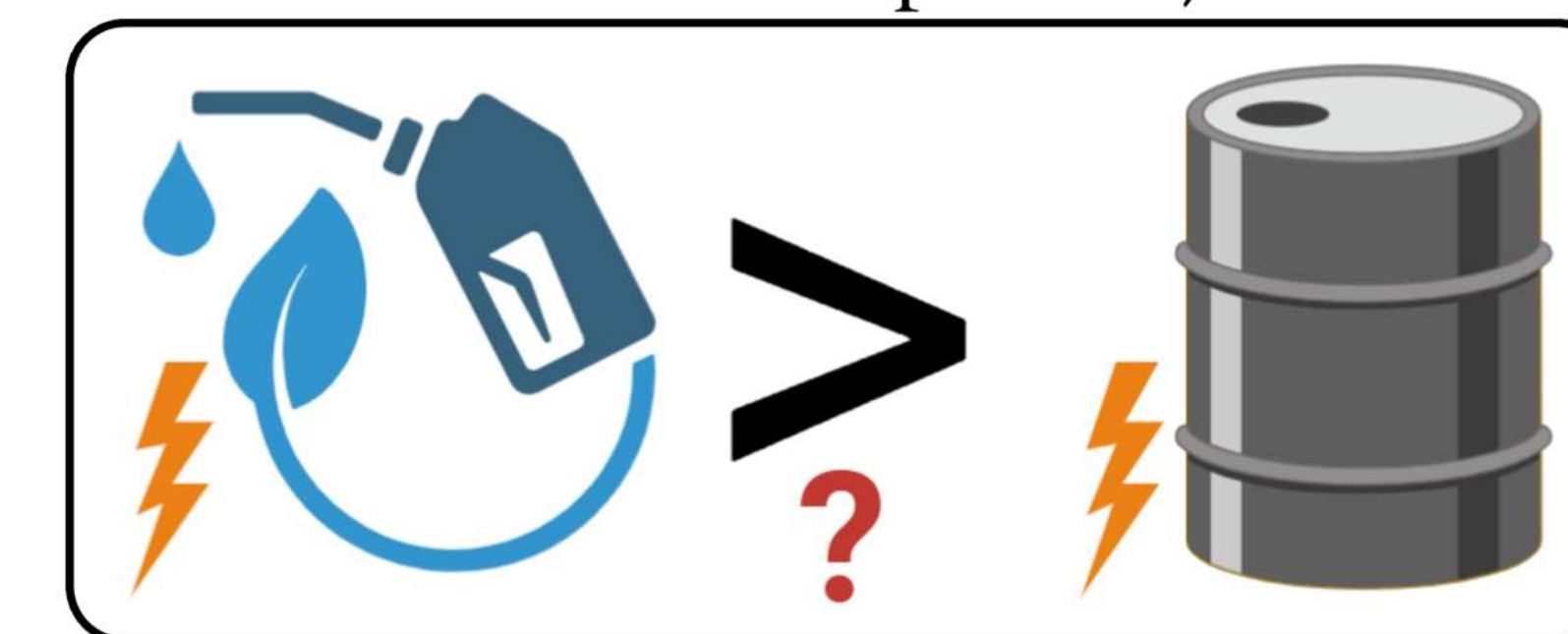
The conversion yield of useful compounds from HDO using Pd/AIO **significantly outperformed** Pd/C (88% vs 11%). The Monte-Carlo simulations afterward confirmed this discrepancy maintained when accounting for costs, showing that the **break-even point** for a liter of compound in Pd/AIO is **significantly less than** Pd/C, by a factor of about 7 (\$65 vs \$480). This suggests that this methodology using Pd/AIO is much more promising than Pd/C but still has a long way to go in technological advancement before it can be comparable with oil compounds.



Graphic created by researchers with BioRender

Future Work

More work should look into **how far HDO technology must come** for the price per liter of products to **rival regular gasoline** compounds by performing several Monte-Carlo simulations for different scenarios, showing how much **investment** would be required to bring costs down to a **competitive margin**. Future work should also analyze the **actual energy derivement** from the HDO compounds (i.e. find how much energy the HDO compounds have *and* how that compares with oil compounds).



Graphic created by researchers with BioRender

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